NICE Roles (Purple + Green + Red) - Tasks and Knowledge Reference

Source workbook: MedData Solutions Roles Nice Framework.xlsx

This report includes all selected NICE roles used in the MedData workforce plan: purple training roles, green internal roles, and the red new-hire role. For each role, the associated Task (T) and Knowledge (K) statements are listed. Skills (S) are intentionally omitted to keep the deliverable focused.

Role Summary

Color Group	Work Role ID	Work Role	Tasks (T)	Knowledge (K)	Total (T+K)
Purple (Training)	OG-WRL-002	Cybersecurity Policy and Planning	25	34	59
Purple (Training)	OG-WRL-008	Privacy Compliance	78	48	126
Purple (Training)	OG-WRL-012	Security Control Assessment	52	132	184
Purple (Training)	OG-WRL-016	Technology Program Auditing	19	38	57
Purple (Training)	PD-WRL-003	Incident Response	29	55	84
Purple (Training)	PD-WRL-004	Infrastructure Support	14	51	65
Purple (Training)	PD-WRL-006	Threat Analysis	38	76	114
Purple (Training)	PD-WRL-007	Vulnerability Analysis	15	64	79
Red (New Hire)	DD-WRL-001	Cybersecurity Architecture	50	121	171
Green (Internal)	IO-WRL-002	Database Administration	14	57	71
Green (Internal)	IO-WRL-005	Systems Administration	26	79	105
Green (Internal)	PD-WRL-001	Defensive Cybersecurity	43	125	168

OG-WRL-002 - Cybersecurity Policy and Planning

Group: Purple (Training)

Responsible for developing and maintaining cybersecurity plans, strategy, and policy to support and align with organizational cybersecurity initiatives and regulatory compliance.

Tasks (T) - 25

Task ID	Task Statement
T0226	Serve on agency and interagency policy boards
T1020	Determine the operational and safety impacts of cybersecurity lapses
T1028	Research new vulnerabilities in emerging technologies
T1107	Evaluate functional requirements
T1158	Develop cybersecurity implementation policies and guidelines
T1184	Establish stakeholder communication channels
T1185	Maintain stakeholder communication channels
T1306	Conduct technology program and project audits
T1335	Promote cybersecurity awareness to management
T1336	Verify the inclusion of sound cybersecurity principles in the organization's vision and goals
T1357	Determine if cybersecurity requirements have been successfully implemented
T1358	Determine the effectiveness of organizational cybersecurity policies and procedures
T1394	Develop independent cybersecurity audit processes for application software, networks, and systems
T1395	Implement independent cybersecurity audit processes for application software, networks, and systems
T1396	Oversee independent cybersecurity audits
T1397	Determine if research and design processes and procedures are in compliance with cybersecurity requirements
T1398	Determine if research and design processes and procedures are accurately followed by cybersecurity staff when performing their day-to-day activities
T1436	Acquire adequate funding for cybersecurity training
T1464	Determine if cybersecurity workforce management policies and procedures comply with legal and organizational requirements
T1476	Promote awareness of cybersecurity policy and strategy among management
T1482	Conduct cybersecurity workforce assessments
T1492	Integrate laws and regulations into policy
T1518	Develop organizational cybersecurity strategy
T1543	Develop cybersecurity policies and procedures
T1605	Advise management, staff, and users on cybersecurity policy

Knowledge (K) - 34

Knowledge ID	Knowledge Statement
K0644	Knowledge of cybersecurity operation policies and procedures
K0674	Knowledge of computer networking protocols
K0675	Knowledge of risk management processes
K0676	Knowledge of cybersecurity laws and regulations
K0677	Knowledge of cybersecurity policies and procedures
K0678	Knowledge of privacy laws and regulations
K0679	Knowledge of privacy policies and procedures
K0680	Knowledge of cybersecurity principles and practices
K0681	Knowledge of privacy principles and practices
K0682	Knowledge of cybersecurity threats
K0683	Knowledge of cybersecurity vulnerabilities
K0684	Knowledge of cybersecurity threat characteristics
K0691	Knowledge of cyber defense tools and techniques
K0692	Knowledge of vulnerability assessment tools and techniques
K0743	Knowledge of new and emerging technologies
K0751	Knowledge of system threats
K0752	Knowledge of system vulnerabilities
K0773	Knowledge of telecommunications principles and practices

Knowledge ID	Knowledge Statement
K0812	Knowledge of digital communication systems and software
K0892	Knowledge of cyber defense laws and regulations
K0943	Knowledge of industry indicators
K0962	Knowledge of targeting laws and regulations
K0963	Knowledge of exploitation laws and regulations
K0969	Knowledge of cyberattack tools and techniques
K0983	Knowledge of computer networking principles and practices
K0990	Knowledge of cyber operations principles and practices
K1014	Knowledge of network security principles and practices
K1023	Knowledge of network exploitation tools and techniques
K1079	Knowledge of web application security risks
K1137	Knowledge of cybersecurity requirements
K1180	Knowledge of organizational cybersecurity goals and objectives
K1183	Knowledge of organizational cybersecurity policies and procedures
K1186	Knowledge of organizational human resource (HR) policies and procedures
K1206	Knowledge of research and design processes and procedures

Skill (S) - 9

Skill ID	Skill Statement
S0406	Skill in developing policy plans
S0497	Skill in developing client organization profiles
S0515	Skill in identifying partner capabilities
S0519	Skill in detecting exploitation activities
S0687	Skill in performing administrative planning activities
S0712	Skill in evaluating data source quality
S0713	Skill in evaluating information quality
S0729	Skill in preparing plans
S0821	Skill in collaborating with internal and external stakeholders

OG-WRL-008 - Privacy Compliance

Group: Purple (Training)

Responsible for developing and overseeing an organization's privacy compliance program and staff, including establishing and managing privacy-related governance, policy, and incident response needs.

Tasks (T) - 78

Task ID	Task Statement
T0898	Establish an internal privacy audit program
T1014	Determine if security incidents require legal action
T1020	Determine the operational and safety impacts of cybersecurity lapses
T1054	Scope analysis reports to various audiences that accounts for data sharing classification restrictions
T1058	Advise senior management on risk levels and security posture
T1059	Perform cost/benefit analyses of cybersecurity programs, policies, processes, systems, and elements
T1060	Advise senior management on organizational cybersecurity efforts
T1084	Identify anomalous network activity
T1092	Conduct functional and connectivity testing
T1096	Perform privacy impact assessments (PIAs)
T1118	Identify vulnerabilities
T1119	Recommend vulnerability remediation strategies
T1145	Develop strategic plans
T1146	Maintain strategic plans
T1189	Determine if contracts comply with funding, legal, and program requirements
T1224	Determine impact of noncompliance on organizational risk levels
T1225	Determine impact of noncompliance on effectiveness of the enterprise's cybersecurity program
T1279	Prepare audit reports
T1334	Produce cybersecurity instructional materials
T1335	Promote cybersecurity awareness to management
T1336	Verify the inclusion of sound cybersecurity principles in the organization's vision and goals
T1476	Promote awareness of cybersecurity policy and strategy among management
T1489	Correlate incident data
T1492	Integrate laws and regulations into policy
T1549	Evaluate the impact of legal, regulatory, policy, standard, or procedural changes
T1853	Determine if new and existing services comply with privacy and data security obligations
T1854	Develop and maintain privacy and confidentiality consent forms
T1855	Develop and maintain privacy and confidentiality authorization forms
T1856	Integrate civil rights and civil liberties in organizational programs, policies, and procedures
T1857	Integrate privacy considerations in organizational programs, policies, and procedures
T1858	Serve as liaison to regulatory and accrediting bodies
T1859	Register databases with local privacy and data protection authorities
T1860	Promote privacy awareness to management
T1861	Establish organizational Privacy Oversight Committee
T1862	Establish cybersecurity risk assessment processes
T1863	Develop information sharing strategic plans
T1864	Develop organizational information infrastructure
T1865	Implement organizational information infrastructure
T1866	Develop self-disclosure policies and procedures
T1867	Oversee consumer information access rights
T1868	Serve as information privacy liaison to technology system users
T1869	Serve as liaison to information systems department
T1870	Create privacy training materials
T1871	Prepare privacy awareness communications
T1872	Deliver privacy awareness orientations
T1873	Deliver privacy awareness trainings
T1874	Manage organizational participation in public privacy and cybersecurity events
T1875	Prepare privacy program status reports
T1876	Respond to press and other public data security inquiries
T1877	Develop organizational privacy program

Task ID	Task Statement
T1878	Apply sanctions for failure to comply with privacy policies
T1879	Develop sanctions for failure to comply with privacy policies
T1880	Resolve allegations of noncompliance with privacy policies and notice of information practices
T1881	Develop a risk management and compliance framework for privacy
T1882	Determine if projects comply with organizational privacy and data security policies
T1883	Develop organizational privacy policies and procedures
T1884	Establish complaint processes
T1885	Establish mechanisms to track access to protected health information
T1886	Maintain the organizational policy program
T1887	Conduct privacy impact assessments
T1888	Conduct privacy compliance monitoring
T1889	Align cybersecurity and privacy practices in system information security plans
T1890	Determine if protected information releases comply with organizational policies and procedures
T1891	Administer requests for release or disclosure of protected information
T1892	Develop vendor review procedures
T1893	Develop vendor auditing procedures
T1894	Determine if partner and business agreements address privacy requirements and responsibilities
T1895	Provide legal advice for business partner contracts
T1896	Mitigate Personal Identifiable Information (PII) breaches
T1897	Administer action on organizational privacy complaints
T1898	Determine if the organization's privacy program complies with federal and state privacy laws and regulations
T1899	Identify organizational privacy compliance gaps
T1900	Correct organizational privacy compliance gaps
T1901	Manage privacy breaches
T1902	Implement and maintain organizational privacy policies and procedures
T1903	Develop and maintain privacy and confidentiality information notices
T1905	Monitor advancements in information privacy technologies
T1907	Establish organizational risk management strategies

Knowledge (K) - 48

Knowledge ID	Knowledge Statement
K0498	Knowledge of operational planning processes
K0644	Knowledge of cybersecurity operation policies and procedures
K0645	Knowledge of standard operating procedures (SOPs)
K0659	Knowledge of information privacy technologies
K0674	Knowledge of computer networking protocols
K0675	Knowledge of risk management processes
K0676	Knowledge of cybersecurity laws and regulations
K0677	Knowledge of cybersecurity policies and procedures
K0678	Knowledge of privacy laws and regulations
K0679	Knowledge of privacy policies and procedures
K0680	Knowledge of cybersecurity principles and practices
K0681	Knowledge of privacy principles and practices
K0682	Knowledge of cybersecurity threats
K0683	Knowledge of cybersecurity vulnerabilities
K0684	Knowledge of cybersecurity threat characteristics
K0687	Knowledge of business operations standards and best practices
K0718	Knowledge of network communications principles and practices
K0748	Knowledge of Privacy Impact Assessment (PIA) principles and practices
K0751	Knowledge of system threats
K0752	Knowledge of system vulnerabilities
K0773	Knowledge of telecommunications principles and practices
K0792	Knowledge of network configurations
K0881	Knowledge of learning assessment tools and techniques
K0885	Knowledge of instructional design principles and practices
K0886	Knowledge of instructional design models and frameworks
K0892	Knowledge of cyber defense laws and regulations

Knowledge ID	Knowledge Statement
K0915	Knowledge of network architecture principles and practices
K0925	Knowledge of wireless communication tools and techniques
K0926	Knowledge of signal jamming tools and techniques
K0962	Knowledge of targeting laws and regulations
K0963	Knowledge of exploitation laws and regulations
K0973	Knowledge of system persistence tools and techniques
K0983	Knowledge of computer networking principles and practices
K0990	Knowledge of cyber operations principles and practices
K1014	Knowledge of network security principles and practices
K1030	Knowledge of operational planning tools and techniques
K1070	Knowledge of privacy disclosure statement laws and regulations
K1111	Knowledge of application security design principles and practices
K1120	Knowledge of Confidentiality, Integrity, Availability, Authenticity, and Non-repudiation (CIAAN) principles and practices
K1138	Knowledge of cybersecurity standards and best practices
K1160	Knowledge of federal and state accreditation standards
K1183	Knowledge of organizational cybersecurity policies and procedures
K1192	Knowledge of organizational privacy policies and procedures
K1194	Knowledge of Personally Identifiable Information (PII) attributes
K1198	Knowledge of privacy and data security regulators
K1200	Knowledge of privacy technologies
K1212	Knowledge of security controls
K1240	Knowledge of privacy laws and regulations

Skill (S) - 21

Skill ID	Skill Statement
S0395	Skill in developing instructional materials
S0406	Skill in developing policy plans
S0407	Skill in developing standard operating procedures (SOPs)
S0408	Skill in maintaining standard operating procedures (SOPs)
S0447	Skill in aligning privacy and cybersecurity objectives
S0450	Skill in authoring privacy disclosure statements
S0537	Skill in designing wireless communications systems
S0540	Skill in identifying network threats
S0601	Skill in developing curricula
S0602	Skills in teaching training programs
S0610	Skill in communicating effectively
S0687	Skill in performing administrative planning activities
S0791	Skill in presenting to an audience
S0796	Skill in creating privacy policies
S0797	Skill in negotiating vendor agreements
S0798	Skill in evaluating vendor privacy practices
S0818	Skill in building internal and external stakeholder relationships
S0821	Skill in collaborating with internal and external stakeholders
S0850	Skill in performing cost/benefit analysis
S0858	Skill in performing economic analysis
S0878	Skill in performing risk analysis

OG-WRL-012 - Security Control Assessment

Group: Purple (Training)

Responsible for conducting independent comprehensive assessments of management, operational, and technical security controls and control enhancements employed within or inherited by a system to determine their overall effectiveness.

Tasks (T) - 52

Task ID	Task Statement
T0309	Assess the effectiveness of security controls
T0495	Manage Accreditation Packages (e.g., ISO/IEC 15026-2)
T1012	Expand network access
T1013	Conduct technical exploitation of a target
T1019	Determine special needs of cyber-physical systems
T1020	Determine the operational and safety impacts of cybersecurity lapses
T1021	Review cyber defense service provider reporting structure
T1022	Review enterprise information technology (IT) goals and objectives
T1023	Identify critical technology procurement requirements
T1026	Determine procurement requirements
T1030	Estimate the impact of collateral damage
T1036	Integrate leadership priorities
T1038	Integrate organization objectives in intelligence collection
T1041	Determine impact of software configurations
T1046	Assess operation performance
T1047	Assess operation impact
T1054	Scope analysis reports to various audiences that accounts for data sharing classification restrictions
T1055	Determine if priority information requirements are satisfied
T1079	Develop cybersecurity risk profiles
T1084	Identify anomalous network activity
T1118	Identify vulnerabilities
T1119	Recommend vulnerability remediation strategies
T1232	Approve accreditation packages
T1263	Perform security reviews
T1264	Identify gaps in security architecture
T1265	Develop a cybersecurity risk management plan
T1266	Recommend risk mitigation strategies
T1269	Conduct risk analysis of applications and systems undergoing major changes
T1270	Plan security authorization reviews for system and network installations
T1271	Conduct security authorization reviews for system and network installations
T1272	Develop security assurance cases for system and network installations
T1294	Advise on Risk Management Framework process activities and documentation
T1305	Determine if authorization and assurance documents identify an acceptable level of risk for software applications, systems, and networks
T1327	Update security documentation to reflect current application and system security design features
T1328	Verify implementation of software, network, and system cybersecurity postures
T1329	Document software, network, and system deviations from implemented security postures
T1330	Recommend required actions to correct software, network, and system deviations from implemented security postures
T1339	Develop cybersecurity compliance processes for external services
T1340	Develop cybersecurity audit processes for external services
T1343	Provide cybersecurity guidance to organizational risk governance processes
T1355	Determine if vulnerability remediation plans are in place
T1356	Develop vulnerability remediation plans
T1357	Determine if cybersecurity requirements have been successfully implemented
T1358	Determine the effectiveness of organizational cybersecurity policies and procedures
T1361	Determine the impact of new system and interface implementations on organization's cybersecurity posture
T1362	Document impact of new system and interface implementations on organization's cybersecurity posture
T1365	Document cybersecurity design and development activities
T1368	Support cybersecurity compliance activities
T1369	Determine if acquisitions, procurement, and outsourcing efforts address cybersecurity requirements

Task ID	Task Statement
T1437	Determine effectiveness of configuration management processes
T1489	Correlate incident data
T1829	Evaluate locally developed tools

Knowledge (K) - 132

Knowledge ID	Knowledge Statement
K0018	Knowledge of encryption algorithms
K0476	Knowledge of language processing tools and techniques
K0653	Knowledge of cybersecurity practices in the acquisition process
K0655	Knowledge of intelligence fusion
K0658	Knowledge of cognitive biases
K0659	Knowledge of information privacy technologies
K0674	Knowledge of computer networking protocols
K0675	Knowledge of risk management processes
K0676	Knowledge of cybersecurity laws and regulations
K0677	Knowledge of cybersecurity policies and procedures
K0678	Knowledge of privacy laws and regulations
K0679	Knowledge of privacy policies and procedures
K0680	Knowledge of cybersecurity principles and practices
K0681	Knowledge of privacy principles and practices
K0682	Knowledge of cybersecurity threats
K0683	Knowledge of cybersecurity vulnerabilities
K0684	Knowledge of cybersecurity threat characteristics
K0685	Knowledge of access control principles and practices
K0686	Knowledge of authentication and authorization tools and techniques
K0687	Knowledge of business operations standards and best practices
K0688	Knowledge of common application vulnerabilities
K0689	Knowledge of network infrastructure principles and practices
K0691	Knowledge of cyber defense tools and techniques
K0692	Knowledge of vulnerability assessment tools and techniques
K0698	Knowledge of cryptographic key management principles and practices
K0701	Knowledge of data backup and recovery policies and procedures
K0702	Knowledge of data warehousing principles and practices
K0703	Knowledge of data mining principles and practices
K0707	Knowledge of database systems and software
K0709	Knowledge of business continuity and disaster recovery (BCDR) policies and procedures
K0710	Knowledge of enterprise cybersecurity architecture principles and practices
K0711	Knowledge of evaluation and validation principles and practices
K0712	Knowledge of Local Area Networks (LAN)
K0713	Knowledge of Wide Area Networks (WAN)
K0718	Knowledge of network communications principles and practices
K0720	Knowledge of Security Assessment and Authorization (SA&A) processes
K0721	Knowledge of risk management principles and practices
K0723	Knowledge of vulnerability data sources
K0728	Knowledge of Confidentiality, Integrity and Availability (CIA) principles and practices
K0729	Knowledge of non-repudiation principles and practices
K0730	Knowledge of cyber safety principles and practices
K0731	Knowledge of systems security engineering (SSE) principles and practices
K0734	Knowledge of Risk Management Framework (RMF) requirements
K0735	Knowledge of risk management models and frameworks
K0736	Knowledge of information technology (IT) security principles and practices
K0742	Knowledge of identity and access management (IAM) principles and practices
K0743	Knowledge of new and emerging technologies
K0746	Knowledge of policy-based access controls
K0747	Knowledge of Risk Adaptive (Adaptable) Access Controls (RAdAC)
K0749	Knowledge of process engineering principles and practices
K0751	Knowledge of system threats

Knowledge ID	Knowledge Statement
K0752	Knowledge of system vulnerabilities
K0758	Knowledge of server administration principles and practices
K0760	Knowledge of server diagnostic tools and techniques
K0761	Knowledge of Fault Detection and Diagnostics (FDD) tools and techniques
K0765	Knowledge of software engineering principles and practices
K0767	Knowledge of structured analysis principles and practices
K0776	Knowledge of collaboration tools and techniques
K0778	Knowledge of enterprise information technology (IT) architecture principles and practices
K0779	Knowledge of systems engineering processes
K0784	Knowledge of insider threat laws and regulations
K0785	Knowledge of insider threat tools and techniques
K0791	Knowledge of defense-in-depth principles and practices
K0800	Knowledge of evidence admissibility laws and regulations
K0803	Knowledge of supply chain risk management principles and practices
K0806	Knowledge of machine virtualization tools and techniques
K0814	Knowledge of secure coding tools and techniques
K0819	Knowledge of import and export control laws and regulations
K0820	Knowledge of supply chain risks
K0821	Knowledge of federal agency roles and responsibilities
K0828	Knowledge of supply chain risk management standards and best practices
K0834	Knowledge of technology procurement principles and practices
K0838	Knowledge of supply chain risk management policies and procedures
K0839	Knowledge of critical infrastructure systems and software
K0840	Knowledge of hardware reverse engineering tools and techniques
K0842	Knowledge of software reverse engineering tools and techniques
K0851	Knowledge of reverse engineering principles and practices
K0858	Knowledge of virtual machine detection tools and techniques
K0859	Knowledge of encryption tools and techniques
K0865	Knowledge of data classification standards and best practices
K0866	Knowledge of data classification tools and techniques
K0870	Knowledge of enterprise architecture (EA) reference models and frameworks
K0871	Knowledge of enterprise architecture (EA) principles and practices
K0877	Knowledge of application firewall principles and practices
K0878	Knowledge of network firewall principles and practices
K0879	Knowledge of industry cybersecurity models and frameworks
K0880	Knowledge of access control models and frameworks
K0881	Knowledge of learning assessment tools and techniques
K0885	Knowledge of instructional design principles and practices
K0886	Knowledge of instructional design models and frameworks
K0892	Knowledge of cyber defense laws and regulations
K0915	Knowledge of network architecture principles and practices
K0917	Knowledge of Personally Identifiable Information (PII) data security standards and best practices
K0918	Knowledge of Payment Card Industry (PCI) data security standards and best practices
K0919	Knowledge of Personal Health Information (PHI) data security standards and best practices
K0922	Knowledge of the acquisition life cycle models and frameworks
K0924	Knowledge of network analysis tools and techniques
K0928	Knowledge of systems engineering principles and practices
K0934	Knowledge of data classification policies and procedures
K0942	Knowledge of cryptology principles and practices
K0947	Knowledge of computer engineering principles and practices
K0948	Knowledge of embedded systems and software
K0953	Knowledge of data mining tools and techniques
K0955	Knowledge of penetration testing principles and practices
K0956	Knowledge of penetration testing tools and techniques
K0962	Knowledge of targeting laws and regulations
K0963	Knowledge of exploitation laws and regulations
K0965	Knowledge of language analysis tools and techniques
K0966	Knowledge of voice analysis tools and techniques
K0967	Knowledge of graphic materials analysis tools and techniques
K0983	Knowledge of computer networking principles and practices

Knowledge ID	Knowledge Statement
K0986	Knowledge of target selection criticality factors
K0987	Knowledge of target selection vulnerability factors
K0990	Knowledge of cyber operations principles and practices
K1014	Knowledge of network security principles and practices
K1034	Knowledge of target language
K1050	Knowledge of critical information requirements
K1063	Knowledge of operation assessment processes
K1069	Knowledge of virtual machine tools and technologies
K1076	Knowledge of risk scoring principles and practices
K1077	Knowledge of data security controls
K1079	Knowledge of web application security risks
K1084	Knowledge of data privacy controls
K1088	Knowledge of knowledge management tools and techniques
K1096	Knowledge of data analysis tools and techniques
K1098	Knowledge of personnel systems and software
K1099	Knowledge of code analysis tools and techniques
K1100	Knowledge of analytical tools and techniques
K1101	Knowledge of analytics
K1108	Knowledge of traceroute tools and techniques
K1109	Knowledge of virtual collaborative workspace tools and techniques
K1180	Knowledge of organizational cybersecurity goals and objectives

Skill (S) - 123

Skill ID	Skill Statement
S0015	Skill in conducting test events
S0097	Skill in applying security controls
S0111	Skill in interfacing with customers
S0136	Skill in network systems management principles, models, methods (e.g., end-to-end systems performance monitoring), and tools
S0141	Skill in assessing security systems designs
S0172	Skill in applying secure coding techniques
S0175	Skill in performing root cause analysis
S0177	Skill in performing network analysis on targets
S0248	Skill in performing target system analysis
S0252	Skill in processing collected data for follow-on analysis
S0385	Skill in communicating complex concepts
S0386	Skill in communicating verbally
S0387	Skill in communicating in writing
S0388	Skill in facilitating small group discussions
S0389	Skill in facilitating group discussions
S0391	Skill in creating technical documentation
S0393	Skill in developing assessments
S0394	Skill in developing security assessments
S0401	Skill in collecting data
S0402	Skill in verifying data
S0403	Skill in validating data
S0409	Skill in deriving evaluative conclusions from data
S0414	Skill in evaluating laws
S0415	Skill in evaluating regulations
S0416	Skill in evaluating policies
S0423	Skill in analyzing processes to ensure conformance with procedural requirements
S0430	Skill in collaborating with others
S0431	Skill in applying critical thinking
S0435	Skill in analyzing large data sets
S0436	Skill in creating target intelligence products
S0437	Skill in identifying targets of interest
S0438	Skill in functioning effectively in a dynamic, fast-paced environment

S0439	Skill in identifying external partners
S0440	Skill in identifying target vulnerabilities
S0441	Skill in describing target vulnerabilities
S0443	Skill in mitigating cognitive biases
S0447	Skill in aligning privacy and cybersecurity objectives
S0462	Skill in integrating information security requirements in the acquisitions process
S0463	Skill in implementing software quality control processes
S0465	Skill in identifying critical infrastructure systems
S0466	Skill in identifying systems designed without security considerations
S0472	Skill in developing virtual machines
S0473	Skill in maintaining virtual machines
S0483	Skill in identifying software communications vulnerabilities
S0503	Skill in selecting targets
S0504	Skill in identifying vulnerabilities
S0506	Skill in identifying customer information needs
S0511	Skill in establishing priorities
S0515	Skill in identifying partner capabilities
S0532	Skill in analyzing software configurations
S0543	Skill in scanning for vulnerabilities
S0544	Skill in recognizing vulnerabilities
S0558	Skill in developing algorithms
S0559	Skill in performing data structure analysis
S0574	Skill in developing security system controls
S0578	Skill in evaluating security designs
S0579	Skill in preparing reports
S0580	Skill in monitoring system performance
S0581	Skill in configuring systems for performance enhancement
S0601	Skill in developing curricula
S0602	Skill in teaching training programs
S0614	Skill in categorizing types of vulnerabilities
S0628	Skill in developing learning activities
S0632	Skill in designing Test and Evaluation Strategies (TES)
S0634	Skill in identifying Test and Evaluation Strategies (TES) infrastructure requirements
S0635	Skill in managing test assets
S0641	Skill in reviewing logs
S0642	Skill in identifying evidence of past intrusions
S0645	Skill in troubleshooting cyber defense infrastructure anomalies
S0647	Skill in managing a workforce
S0654	Skill in conducting system reviews
S0655	Skill in designing secure test plans
S0656	Skill in assessing application vulnerabilities
S0657	Skill in implementing Public Key Infrastructure (PKI) encryption
S0658	Skill in implementing digital signatures
S0664	Skill in applying policies that meet system security objectives
S0667	Skill in assessing security controls
S0673	Skill in translating operational requirements into security controls
S0675	Skill in optimizing system performance
S0686	Skill in performing risk assessments
S0687	Skill in performing administrative planning activities
S0688	Skill in performing network data analysis
S0700	Skill in mining data
S0701	Skill in performing data mining analysis
S0704	Skill in performing target analysis
S0709	Skill in developing analytics
S0710	Skill in evaluating metadata
S0711	Skill in interpreting metadata
S0712	Skill in evaluating data source quality
S0713	Skill in evaluating information quality
S0715	Skill in generating operation plans

S0719	Skill in identifying intelligence gaps
S0720	Skill in identifying regional languages and dialects
S0721	Skill in prioritizing information
S0722	Skill in interpreting traceroute results
S0723	Skill in interpreting vulnerability scanner results
S0724	Skill in managing client relationships
S0728	Skill in preparing briefings
S0729	Skill in preparing plans
S0731	Skill in producing after-action reports
S0739	Skill in analyzing intelligence products
S0743	Skill in identifying network anomalies
S0744	Skill in performing technical writing
S0755	Skill in reconstructing a network
S0756	Skill in incorporating feedback
S0758	Skill in performing wireless network analysis
S0760	Skill in navigating databases
S0761	Skill in performing strategic guidance analysis
S0775	Skill in developing intelligence collection plans
S0777	Skill in developing collection strategies
S0780	Skill in fulfilling information requests
S0788	Skill in orchestrating planning teams
S0789	Skill in coordinating collection support
S0790	Skill in monitoring status
S0791	Skill in presenting to an audience
S0800	Skill in analyzing organizational patterns and relationships
S0801	Skill in assessing partner operations capabilities
S0807	Skill in solving problems
S0809	Skill in utilizing cyber defense service provider information
S0813	Skill in identifying cybersecurity issues in external connections
S0814	Skill in identifying privacy issues in partner interconnections
S0874	Skill in performing network traffic analysis
S0878	Skill in performing risk analysis

OG-WRL-016 - Technology Program Auditing

Group: Purple (Training)

Responsible for conducting evaluations of technology programs or their individual components to determine compliance with published standards.

Tasks (T) - 19

Task ID	Task Statement
T0412	Conduct import/export reviews for acquiring systems and software
T1020	Determine the operational and safety impacts of cybersecurity lapses
T1026	Determine procurement requirements
T1067	Recommend development of new applications or modification of existing applications
T1068	Create development plans for new applications or modification of existing applications
T1154	Develop risk, compliance, and assurance monitoring strategies
T1155	Develop risk, compliance, and assurance measurement strategies
T1306	Conduct technology program and project audits
T1344	Determine if procurement activities sufficiently address supply chain risks
T1345	Recommend improvements to procurement activities to address cybersecurity requirements
T1394	Develop independent cybersecurity audit processes for application software, networks, and systems
T1395	Implement independent cybersecurity audit processes for application software, networks, and systems
T1396	Oversee independent cybersecurity audits
T1397	Determine if research and design processes and procedures are in compliance with cybersecurity requirements
T1398	Determine if research and design processes and procedures are accurately followed by cybersecurity staff when performing their day-to-day activities
T1399	Develop supply chain, system, network, and operational security contract language
T1480	Examine service performance reports for issues and variances
T1481	Initiate corrective actions to service performance issues and variances
T1498	Determine if cybersecurity requirements included in contracts are delivered

Knowledge (K) - 38

Knowledge ID	Knowledge Statement
K0653	Knowledge of cybersecurity practices in the acquisition process
K0674	Knowledge of computer networking protocols
K0675	Knowledge of risk management processes
K0676	Knowledge of cybersecurity laws and regulations
K0677	Knowledge of cybersecurity policies and procedures
K0678	Knowledge of privacy laws and regulations
K0679	Knowledge of privacy policies and procedures
K0680	Knowledge of cybersecurity principles and practices
K0681	Knowledge of privacy principles and practices
K0682	Knowledge of cybersecurity threats
K0683	Knowledge of cybersecurity vulnerabilities
K0684	Knowledge of cybersecurity threat characteristics
K0721	Knowledge of risk management principles and practices
K0727	Knowledge of analysis standards and best practices
K0733	Knowledge of information technology (IT) architecture models and frameworks
K0734	Knowledge of Risk Management Framework (RMF) requirements
K0735	Knowledge of risk management models and frameworks
K0751	Knowledge of system threats
K0752	Knowledge of system vulnerabilities
K0754	Knowledge of resource management principles and practices
K0771	Knowledge of system life cycle management principles and practices
K0803	Knowledge of supply chain risk management principles and practices
K0819	Knowledge of import and export control laws and regulations
K0820	Knowledge of supply chain risks
K0821	Knowledge of federal agency roles and responsibilities

Knowledge ID	Knowledge Statement
K0828	Knowledge of supply chain risk management standards and best practices
K0835	Knowledge of risk assessment principles and practices
K0836	Knowledge of threat assessment principles and practices
K0838	Knowledge of supply chain risk management policies and procedures
K0868	Knowledge of process improvement principles and practices
K0869	Knowledge of process maturity models and frameworks
K0872	Knowledge of service management principles and practices
K0873	Knowledge of service management standards and best practices
K0922	Knowledge of the acquisition life cycle models and frameworks
K0983	Knowledge of computer networking principles and practices
K1014	Knowledge of network security principles and practices
K1137	Knowledge of cybersecurity requirements
K1206	Knowledge of research and design processes and procedures

Skill (S) - 7

Skill ID	Skill Statement
S0423	Skill in analyzing processes to ensure conformance with procedural requirements
S0462	Skill in integrating information security requirements in the acquisitions process
S0463	Skill in implementing software quality control processes
S0580	Skill in monitoring system performance
S0581	Skill in configuring systems for performance enhancement
S0619	Skill in auditing technical systems
S0811	Skill in managing intelligence collection requirements

PD-WRL-003 - Incident Response

Group: Purple (Training)

Responsible for investigating, analyzing, and responding to network cybersecurity incidents.

Tasks (T) - 29

Task ID	Task Statement
T0164	Perform cyber defense trend analysis and reporting
T0262	Employ approved defense-in-depth principles and practices (e.g., defense-in-multiple places, layered defenses, security robustness)
T0510	Coordinate incident response functions
T1020	Determine the operational and safety impacts of cybersecurity lapses
T1084	Identify anomalous network activity
T1085	Identify potential threats to network resources
T1109	Resolve cyber defense incidents
T1110	Coordinate technical support to enterprise-wide cybersecurity defense technicians
T1118	Identify vulnerabilities
T1119	Recommend vulnerability remediation strategies
T1250	Perform cyber defense incident triage
T1251	Recommend incident remediation strategies
T1252	Determine the scope, urgency, and impact of cyber defense incidents
T1256	Perform forensically sound image collection
T1257	Recommend mitigation and remediation strategies for enterprise systems
T1260	Perform real-time cyber defense incident handling
T1299	Determine causes of network alerts
T1315	Track cyber defense incidents from initial detection through final resolution
T1316	Document cyber defense incidents from initial detection through final resolution
T1332	Produce incident findings reports
T1333	Communicate incident findings to appropriate constituencies
T1370	Collect intrusion artifacts
T1371	Mitigate potential cyber defense incidents
T1372	Advise law enforcement personnel as technical expert
T1407	Correlate threat assessment data
T1485	Prepare after action reviews (AARs)
T1489	Correlate incident data
T1582	Maintain currency of cyber defense threat conditions
T1617	Prepare cyber defense reports

Knowledge (K) - 55

Knowledge ID	Knowledge Statement
K0674	Knowledge of computer networking protocols
K0675	Knowledge of risk management processes
K0676	Knowledge of cybersecurity laws and regulations
K0677	Knowledge of cybersecurity policies and procedures
K0678	Knowledge of privacy laws and regulations
K0679	Knowledge of privacy policies and procedures
K0680	Knowledge of cybersecurity principles and practices
K0681	Knowledge of privacy principles and practices
K0682	Knowledge of cybersecurity threats
K0683	Knowledge of cybersecurity vulnerabilities
K0684	Knowledge of cybersecurity threat characteristics
K0685	Knowledge of access control principles and practices
K0686	Knowledge of authentication and authorization tools and techniques
K0689	Knowledge of network infrastructure principles and practices
K0701	Knowledge of data backup and recovery policies and procedures
K0709	Knowledge of business continuity and disaster recovery (BCDR) policies and procedures
K0710	Knowledge of enterprise cybersecurity architecture principles and practices
K0716	Knowledge of host access control (HAC) systems and software

Knowledge ID	Knowledge Statement
K0717	Knowledge of network access control (NAC) systems and software
K0718	Knowledge of network communications principles and practices
K0724	Knowledge of incident response principles and practices
K0725	Knowledge of incident response tools and techniques
K0726	Knowledge of incident handling tools and techniques
K0732	Knowledge of intrusion detection tools and techniques
K0746	Knowledge of policy-based access controls
K0747	Knowledge of Risk Adaptive (Adaptable) Access Controls (RAdAC)
K0751	Knowledge of system threats
K0752	Knowledge of system vulnerabilities
K0770	Knowledge of system administration principles and practices
K0778	Knowledge of enterprise information technology (IT) architecture principles and practices
K0783	Knowledge of network attack characteristics
K0791	Knowledge of defense-in-depth principles and practices
K0829	Knowledge of account creation policies and procedures
K0830	Knowledge of password policies and procedures
K0832	Knowledge of cyberattack characteristics
K0833	Knowledge of cyberattack actor characteristics
K0837	Knowledge of hardening tools and techniques
K0844	Knowledge of cyberattack stages
K0845	Knowledge of cyber intrusion activity phases
K0857	Knowledge of malware analysis tools and techniques
K0865	Knowledge of data classification standards and best practices
K0866	Knowledge of data classification tools and techniques
K0870	Knowledge of enterprise architecture (EA) reference models and frameworks
K0871	Knowledge of enterprise architecture (EA) principles and practices
K0891	Knowledge of the Open Systems Interconnect (OSI) reference model
K0898	Knowledge of cloud service models and frameworks
K0915	Knowledge of network architecture principles and practices
K0916	Knowledge of malware analysis principles and practices
K0924	Knowledge of network analysis tools and techniques
K0934	Knowledge of data classification policies and procedures
K0969	Knowledge of cyberattack tools and techniques
K0983	Knowledge of computer networking principles and practices
K1014	Knowledge of network security principles and practices
K1049	Knowledge of routing protocols
K1079	Knowledge of web application security risks

Skill (S) - 23

Skill ID	Skill Statement
S0077	Skill in securing network communications
S0080	Skill in performing damage assessments
S0483	Skill in identifying software communications vulnerabilities
S0509	Skill in evaluating security products
S0544	Skill in recognizing vulnerabilities
S0547	Skill in identifying malware
S0548	Skill in capturing malware
S0549	Skill in containing malware
S0550	Skill in reporting malware
S0572	Skill in detecting host- and network-based intrusions
S0589	Skill in preserving digital evidence integrity
S0607	Skill in collecting digital evidence
S0608	Skill in processing digital evidence
S0609	Skill in transporting digital evidence
S0614	Skill in categorizing types of vulnerabilities
S0615	Skill in protecting a network against malware
S0651	Skill in performing malware analysis
S0688	Skill in performing network data analysis

S0805	Skill in designing incident responses
S0806	Skill in performing incident responses
S0821	Skill in collaborating with internal and external stakeholders
S0854	Skill in performing data analysis
S0866	Skill in performing log file analysis

PD-WRL-004 - Infrastructure Support

Group: Purple (Training)

Responsible for testing, implementing, deploying, maintaining, and administering infrastructure hardware and software for cybersecurity.

Tasks (T) - 14

Task ID	Task Statement
T1020	Determine the operational and safety impacts of cybersecurity lapses
T1111	Administer rule and signature updates for specialized cyber defense applications
T1267	Perform system administration on specialized cyber defense applications and systems
T1268	Administer Virtual Private Network (VPN) devices
T1352	Coordinate critical cyber defense infrastructure protection measures
T1353	Prioritize critical cyber defense infrastructure resources
T1432	Build dedicated cyber defense hardware
T1433	Install dedicated cyber defense hardware
T1442	Assess the impact of implementing and sustaining a dedicated cyber defense infrastructure
T1503	Evaluate platforms managed by service providers
T1515	Manage network access control lists on specialized cyber defense systems
T1555	Implement cyber defense tools
T1561	Implement dedicated cyber defense systems
T1562	Document system requirements

Knowledge (K) - 51

Knowledge ID	Knowledge Statement
K0674	Knowledge of computer networking protocols
K0675	Knowledge of risk management processes
K0676	Knowledge of cybersecurity laws and regulations
K0677	Knowledge of cybersecurity policies and procedures
K0678	Knowledge of privacy laws and regulations
K0679	Knowledge of privacy policies and procedures
K0680	Knowledge of cybersecurity principles and practices
K0681	Knowledge of privacy principles and practices
K0682	Knowledge of cybersecurity threats
K0683	Knowledge of cybersecurity vulnerabilities
K0684	Knowledge of cybersecurity threat characteristics
K0685	Knowledge of access control principles and practices
K0686	Knowledge of authentication and authorization tools and techniques
K0701	Knowledge of data backup and recovery policies and procedures
K0710	Knowledge of enterprise cybersecurity architecture principles and practices
K0716	Knowledge of host access control (HAC) systems and software
K0717	Knowledge of network access control (NAC) systems and software
K0724	Knowledge of incident response principles and practices
K0725	Knowledge of incident response tools and techniques
K0726	Knowledge of incident handling tools and techniques
K0728	Knowledge of Confidentiality, Integrity and Availability (CIA) principles and practices
K0729	Knowledge of non-repudiation principles and practices
K0730	Knowledge of cyber safety principles and practices
K0734	Knowledge of Risk Management Framework (RMF) requirements
K0746	Knowledge of policy-based access controls
K0747	Knowledge of Risk Adaptive (Adaptable) Access Controls (RAdAC)
K0751	Knowledge of system threats
K0752	Knowledge of system vulnerabilities
K0770	Knowledge of system administration principles and practices
K0778	Knowledge of enterprise information technology (IT) architecture principles and practices
K0781	Knowledge of virtual private network (VPN) systems and software

Knowledge ID	Knowledge Statement
K0783	Knowledge of network attack characteristics
K0791	Knowledge of defense-in-depth principles and practices
K0792	Knowledge of network configurations
K0811	Knowledge of web filtering systems and software
K0829	Knowledge of account creation policies and procedures
K0830	Knowledge of password policies and procedures
K0837	Knowledge of hardening tools and techniques
K0870	Knowledge of enterprise architecture (EA) reference models and frameworks
K0871	Knowledge of enterprise architecture (EA) principles and practices
K0881	Knowledge of learning assessment tools and techniques
K0891	Knowledge of the Open Systems Interconnect (OSI) reference model
K0915	Knowledge of network architecture principles and practices
K0925	Knowledge of wireless communication tools and techniques
K0926	Knowledge of signal jamming tools and techniques
K0950	Knowledge of Intrusion Detection System (IDS) tools and techniques
K0951	Knowledge of Intrusion Prevention System (IPS) tools and techniques
K0983	Knowledge of computer networking principles and practices
K1014	Knowledge of network security principles and practices
K1211	Knowledge of security assessment authorization requirements
K1237	Knowledge of Virtual Private Network (VPN) devices

Skill (S) – 11

Skill ID	Skill Statement
S0077	Skill in securing network communications
S0552	Skill in applying host access controls
S0553	Skill in applying network access controls
S0592	Skill in tuning network sensors
S0593	Skill in handling incidents
S0596	Skill in encrypting network communications
S0615	Skill in protecting a network against malware
S0643	Skill in applying hardening techniques
S0645	Skill in troubleshooting cyber defense infrastructure anomalies
S0831	Skill in configuring hardware
S0898	Skill in testing hardware

PD-WRL-006 - Threat Analysis

Group: Purple (Training)

Responsible for collecting, processing, analyzing, and disseminating cybersecurity threat assessments. Develops cybersecurity indicators to maintain awareness of the status of the highly dynamic operating environment.

Tasks (T) - 38

Task ID	Task Statement
T0569	Answer requests for information
T0685	Evaluate threat decision-making processes
T0698	Facilitate continuously updated intelligence, surveillance, and visualization input to common operational picture managers
T0707	Generate requests for information
T0718	Identify intelligence gaps and shortfalls
T0751	Monitor open source websites for hostile content directed towards organizational or partner interests
T0845	Identify cyber threat tactics and methodologies
T1020	Determine the operational and safety impacts of cybersecurity lapses
T1035	Determine how threat activity groups employ encryption to support their operations
T1053	Identify and characterize intrusion activities against a victim or target
T1054	Scope analysis reports to various audiences that accounts for data sharing classification restrictions
T1055	Determine if priority information requirements are satisfied
T1640	Determine effectiveness of intelligence collection operations
T1641	Recommend adjustments to intelligence collection strategies
T1643	Develop common operational pictures
T1644	Develop cyber operations indicators
T1645	Coordinate all-source collection activities
T1646	Validate all-source collection requirements and plans
T1647	Develop priority information requirements
T1651	Prepare threat and target briefings
T1652	Prepare threat and target situational updates
T1686	Identify intelligence requirements
T1762	Modify collection requirements
T1763	Determine effectiveness of collection requirements
T1765	Monitor changes to designated cyber operations warning problem sets
T1766	Prepare change reports for designated cyber operations warning problem sets
T1767	Monitor threat activities
T1768	Prepare threat activity reports
T1770	Report on adversarial activities that fulfill priority information requirements
T1772	Identify indications and warnings of target communication changes or processing failures
T1775	Prepare cyber operations intelligence reports
T1776	Prepare indications and warnings intelligence reports
T1792	Assess effectiveness of intelligence production
T1793	Assess effectiveness of intelligence reporting
T1798	Provide intelligence analysis and support
T1799	Notify appropriate personnel of imminent hostile intentions or activities
T1804	Prepare network intrusion reports
T1835	Determine if intelligence requirements and collection plans are accurate and up-to-date

Knowledge (K) - 76

Knowledge ID	Knowledge Statement
K0018	Knowledge of encryption algorithms
K0480	Knowledge of malware
K0655	Knowledge of intelligence fusion
K0658	Knowledge of cognitive biases
K0674	Knowledge of computer networking protocols
K0675	Knowledge of risk management processes

Knowledge ID	Knowledge Statement
K0676	Knowledge of cybersecurity laws and regulations
K0677	Knowledge of cybersecurity policies and procedures
K0678	Knowledge of privacy laws and regulations
K0679	Knowledge of privacy policies and procedures
K0680	Knowledge of cybersecurity principles and practices
K0681	Knowledge of privacy principles and practices
K0682	Knowledge of cybersecurity threats
K0683	Knowledge of cybersecurity vulnerabilities
K0684	Knowledge of cybersecurity threat characteristics
K0689	Knowledge of network infrastructure principles and practices
K0690	Knowledge of requirements analysis principles and practices
K0697	Knowledge of encryption algorithm capabilities and applications
K0718	Knowledge of network communications principles and practices
K0719	Knowledge of human-computer interaction (HCI) principles and practices
K0751	Knowledge of system threats
K0752	Knowledge of system vulnerabilities
K0766	Knowledge of data asset management principles and practices
K0773	Knowledge of telecommunications principles and practices
K0786	Knowledge of physical computer components
K0787	Knowledge of computer peripherals
K0788	Knowledge of adversarial tactics principles and practices
K0789	Knowledge of adversarial tactics tools and techniques
K0790	Knowledge of adversarial tactics policies and procedures
K0792	Knowledge of network configurations
K0806	Knowledge of machine virtualization tools and techniques
K0812	Knowledge of digital communication systems and software
K0818	Knowledge of new and emerging cybersecurity risks
K0825	Knowledge of threat vector characteristics
K0831	Knowledge of network attack vectors
K0844	Knowledge of cyberattack stages
K0845	Knowledge of cyber intrusion activity phases
K0857	Knowledge of malware analysis tools and techniques
K0858	Knowledge of virtual machine detection tools and techniques
K0865	Knowledge of data classification standards and best practices
K0866	Knowledge of data classification tools and techniques
K0915	Knowledge of network architecture principles and practices
K0916	Knowledge of malware analysis principles and practices
K0925	Knowledge of wireless communication tools and techniques
K0926	Knowledge of signal jamming tools and techniques
K0934	Knowledge of data classification policies and procedures
K0960	Knowledge of content management system (CMS) capabilities and applications
K0969	Knowledge of cyberattack tools and techniques
K0983	Knowledge of computer networking principles and practices
K0984	Knowledge of web security principles and practices
K0989	Knowledge of intelligence information repositories
K0990	Knowledge of cyber operations principles and practices
K0994	Knowledge of denial and deception tools and techniques
K1002	Knowledge of supervisory control and data acquisition (SCADA) systems and software
K1005	Knowledge of intelligence collection capabilities and applications
K1007	Knowledge of intelligence requirements tasking systems and software
K1008	Knowledge of intelligence support activities
K1009	Knowledge of threat intelligence principles and practices
K1010	Knowledge of intelligence policies and procedures
K1011	Knowledge of network addressing principles and practices
K1014	Knowledge of network security principles and practices
K1019	Knowledge of operations security (OPSEC) principles and practices
K1025	Knowledge of decision-making policies and procedures
K1028	Knowledge of target development principles and practices
K1035	Knowledge of target research tools and techniques
K1049	Knowledge of routing protocols

Knowledge ID	Knowledge Statement
K1059	Knowledge of request for information processes
K1066	Knowledge of threat behaviors
K1067	Knowledge of target behaviors
K1068	Knowledge of threat systems and software
K1069	Knowledge of virtual machine tools and technologies
K1100	Knowledge of analytical tools and techniques
K1101	Knowledge of analytics
K1109	Knowledge of virtual collaborative workspace tools and techniques
K1113	Knowledge of blue force tracking
K1197	Knowledge of priority intelligence requirements

Skill (S) – 51

Skill ID	Skill Statement
S0111	Skill in interfacing with customers
S0194	Skill in conducting non-attributable research
S0385	Skill in communicating complex concepts
S0430	Skill in collaborating with others
S0433	Skill in creating analytics
S0434	Skill in extrapolating from incomplete data sets
S0435	Skill in analyzing large data sets
S0436	Skill in creating target intelligence products
S0438	Skill in functioning effectively in a dynamic, fast-paced environment
S0443	Skill in mitigating cognitive biases
S0444	Skill in mitigating deception in reporting and analysis
S0446	Skill in mimicking threat actors
S0472	Skill in developing virtual machines
S0473	Skill in maintaining virtual machines
S0494	Skill in performing operational environment analysis
S0505	Skill in performing intrusion data analysis
S0506	Skill in identifying customer information needs
S0509	Skill in evaluating security products
S0511	Skill in establishing priorities
S0512	Skill in extracting metadata
S0514	Skill in preparing operational environments
S0516	Skill in performing threat emulation tactics
S0517	Skill in anticipating threats
S0535	Skill in performing threat factor analysis
S0537	Skill in designing wireless communications systems
S0540	Skill in identifying network threats
S0555	Skill in performing capabilities analysis
S0556	Skill in performing requirements analysis
S0579	Skill in preparing reports
S0600	Skill in collecting relevant data from a variety of sources
S0633	Skill in developing position qualification requirements
S0673	Skill in translating operational requirements into security controls
S0696	Skill in conducting deep web research
S0702	Skill in defining an operational environment
S0704	Skill in performing target analysis
S0709	Skill in developing analytics
S0712	Skill in evaluating data source quality
S0713	Skill in evaluating information quality
S0718	Skill in identifying cybersecurity threats
S0719	Skill in identifying intelligence gaps
S0724	Skill in managing client relationships
S0728	Skill in preparing briefings
S0748	Skill in querying data
S0751	Skill in conducting open-source searches

S0756	Skill in incorporating feedback
S0765	Skill in converting intelligence requirements into intelligence production tasks
S0777	Skill in developing collection strategies
S0779	Skill in determining information requirements
S0791	Skill in presenting to an audience
S0869	Skill in performing metadata analysis
S0876	Skill in performing nodal analysis

PD-WRL-007 - Vulnerability Analysis

Group: Purple (Training)

Responsible for assessing systems and networks to identify deviations from acceptable configurations, enclave policy, or local policy. Measure effectiveness of defense-in-depth architecture against known vulnerabilities.

Tasks (T) - 15

Task ID	Task Statement
T1020	Determine the operational and safety impacts of cybersecurity lapses
T1041	Determine impact of software configurations
T1069	Evaluate organizational cybersecurity policy regulatory compliance
T1070	Evaluate organizational cybersecurity policy alignment with organizational directives
T1079	Develop cybersecurity risk profiles
T1084	Identify anomalous network activity
T1091	Perform authorized penetration testing on enterprise network assets
T1118	Identify vulnerabilities
T1119	Recommend vulnerability remediation strategies
T1229	Maintain deployable cyber defense audit toolkits
T1279	Prepare audit reports
T1341	Perform required reviews
T1489	Correlate incident data
T1619	Perform risk and vulnerability assessments
T1620	Recommend cost-effective security controls

Knowledge (K) - 64

Knowledge ID	Knowledge Statement
K0068	Knowledge of programming language structures and logic
K0674	Knowledge of computer networking protocols
K0675	Knowledge of risk management processes
K0676	Knowledge of cybersecurity laws and regulations
K0677	Knowledge of cybersecurity policies and procedures
K0678	Knowledge of privacy laws and regulations
K0679	Knowledge of privacy policies and procedures
K0680	Knowledge of cybersecurity principles and practices
K0681	Knowledge of privacy principles and practices
K0682	Knowledge of cybersecurity threats
K0683	Knowledge of cybersecurity vulnerabilities
K0684	Knowledge of cybersecurity threat characteristics
K0685	Knowledge of access control principles and practices
K0686	Knowledge of authentication and authorization tools and techniques
K0688	Knowledge of common application vulnerabilities
K0698	Knowledge of cryptographic key management principles and practices
K0701	Knowledge of data backup and recovery policies and procedures
K0710	Knowledge of enterprise cybersecurity architecture principles and practices
K0716	Knowledge of host access control (HAC) systems and software
K0717	Knowledge of network access control (NAC) systems and software
K0728	Knowledge of Confidentiality, Integrity and Availability (CIA) principles and practices
K0729	Knowledge of non-repudiation principles and practices
K0730	Knowledge of cyber safety principles and practices
K0742	Knowledge of identity and access management (IAM) principles and practices
K0751	Knowledge of system threats
K0752	Knowledge of system vulnerabilities
K0760	Knowledge of server diagnostic tools and techniques
K0761	Knowledge of Fault Detection and Diagnostics (FDD) tools and techniques
K0770	Knowledge of system administration principles and practices

Knowledge ID	Knowledge Statement
K0778	Knowledge of enterprise information technology (IT) architecture principles and practices
K0779	Knowledge of systems engineering processes
K0783	Knowledge of network attack characteristics
K0791	Knowledge of defense-in-depth principles and practices
K0797	Knowledge of ethical hacking tools and techniques
K0813	Knowledge of interpreted and compiled programming language characteristics
K0832	Knowledge of cyberattack characteristics
K0833	Knowledge of cyberattack actor characteristics
K0837	Knowledge of hardening tools and techniques
K0844	Knowledge of cyberattack stages
K0845	Knowledge of cyber intrusion activity phases
K0865	Knowledge of data classification standards and best practices
K0866	Knowledge of data classification tools and techniques
K0870	Knowledge of enterprise architecture (EA) reference models and frameworks
K0871	Knowledge of enterprise architecture (EA) principles and practices
K0879	Knowledge of industry cybersecurity models and frameworks
K0880	Knowledge of access control models and frameworks
K0882	Knowledge of ethical hacking principles and practices
K0891	Knowledge of the Open Systems Interconnect (OSI) reference model
K0915	Knowledge of network architecture principles and practices
K0924	Knowledge of network analysis tools and techniques
K0934	Knowledge of data classification policies and procedures
K0939	Knowledge of packet-level analysis tools and techniques
K0942	Knowledge of cryptology principles and practices
K0955	Knowledge of penetration testing principles and practices
K0956	Knowledge of penetration testing tools and techniques
K0969	Knowledge of cyberattack tools and techniques
K0983	Knowledge of computer networking principles and practices
K1014	Knowledge of network security principles and practices
K1076	Knowledge of risk scoring principles and practices
K1079	Knowledge of web application security risks
K1087	Knowledge of social engineering tools and techniques
K1129	Knowledge of cyber defense auditing laws and regulations
K1130	Knowledge of cyber defense auditing policies and practices
K1182	Knowledge of organizational cybersecurity policies and configurations

Skill (S) – 18

Skill ID	Skill Statement
S0483	Skill in identifying software communications vulnerabilities
S0492	Skill in performing threat environment analysis
S0532	Skill in analyzing software configurations
S0543	Skill in scanning for vulnerabilities
S0544	Skill in recognizing vulnerabilities
S0572	Skill in detecting host- and network-based intrusions
S0574	Skill in developing security system controls
S0578	Skill in evaluating security designs
S0588	Skill in performing threat modeling
S0591	Skill in performing social engineering
S0597	Skill in writing code in a currently supported programming language
S0641	Skill in reviewing logs
S0642	Skill in identifying evidence of past intrusions
S0656	Skill in assessing application vulnerabilities
S0675	Skill in optimizing system performance
S0686	Skill in performing risk assessments
S0688	Skill in performing network data analysis
S0804	Skill in assessing an organization's threat environment

DD-WRL-001 - Cybersecurity Architecture

Group: Red (New Hire)

Responsible for ensuring that security requirements are adequately addressed in all aspects of enterprise architecture, including reference models, segment and solution architectures, and the resulting systems that protect and support organizational mission and business processes.

Tasks (T) - 50

Task ID	Task Statement
T0084	Employ secure configuration management processes
T0542	Translate proposed capabilities into technical requirements
T1010	Communicate enterprise information technology architecture
T1019	Determine special needs of cyber-physical systems
T1020	Determine the operational and safety impacts of cybersecurity lapses
T1027	Integrate organizational goals and objectives into security architecture
T1029	Implement organizational evaluation and validation criteria
T1077	Assess the organization's cybersecurity architecture
T1096	Perform privacy impact assessments (PIAs)
T1100	Configure network hubs, routers, and switches
T1101	Optimize network hubs, routers, and switches
T1122	Determine essential system capabilities and business functions
T1123	Prioritize essential system capabilities and business functions
T1124	Restore essential system capabilities and business functions after catastrophic failure events
T1125	Define system availability levels
T1126	Determine disaster recovery and continuity of operations system requirements
T1151	Develop cybersecurity designs for systems and networks with multilevel security requirements
T1152	Develop cybersecurity designs for systems and networks that require processing of multiple data classification levels
T1153	Integrate cybersecurity designs for systems and networks
T1168	Define acquisition life cycle cybersecurity architecture requirements
T1169	Define acquisition life cycle systems security engineering requirements
T1179	Determine if systems and architecture are consistent with cybersecurity architecture guidelines
T1263	Perform security reviews
T1264	Identify gaps in security architecture
T1265	Develop a cybersecurity risk management plan
T1293	Advise on security requirements to be included in statements of work
T1294	Advise on Risk Management Framework process activities and documentation
T1361	Determine the impact of new system and interface implementations on organization's cybersecurity posture
T1362	Document impact of new system and interface implementations on organization's cybersecurity posture
T1363	Plan system security development
T1364	Conduct system security development
T1403	Allocate cybersecurity services
T1404	Select cybersecurity mechanisms
T1410	Develop system security contexts
T1423	Create system security concept of operations (ConOps) documents
T1426	Determine cybersecurity design and architecture effectiveness
T1434	Create cybersecurity architecture functional specifications
T1507	Determine user requirements
T1508	Plan cybersecurity architecture
T1519	Design system security measures
T1520	Update system security measures
T1521	Develop enterprise architecture
T1527	Define baseline system security requirements
T1544	Create definition activity documentation
T1545	Create architecture activity documentation
T1556	Identify system and network protection needs
T1563	Implement system security measures
T1583	Determine effectiveness of system implementation and testing processes
T1627	Conduct cybersecurity management assessments

Task ID	Task Statement
T1628	Design cybersecurity management functions

Knowledge (K) - 121

Knowledge ID	Knowledge Statement
K0018	Knowledge of encryption algorithms
K0055	Knowledge of microprocessors
K0092	Knowledge of technology integration processes
K0646	Knowledge of system optimization techniques
K0674	Knowledge of computer networking protocols
K0675	Knowledge of risk management processes
K0676	Knowledge of cybersecurity laws and regulations
K0677	Knowledge of cybersecurity policies and procedures
K0678	Knowledge of privacy laws and regulations
K0679	Knowledge of privacy policies and procedures
K0680	Knowledge of cybersecurity principles and practices
K0681	Knowledge of privacy principles and practices
K0682	Knowledge of cybersecurity threats
K0683	Knowledge of cybersecurity vulnerabilities
K0684	Knowledge of cybersecurity threat characteristics
K0685	Knowledge of access control principles and practices
K0686	Knowledge of authentication and authorization tools and techniques
K0687	Knowledge of business operations standards and best practices
K0688	Knowledge of common application vulnerabilities
K0689	Knowledge of network infrastructure principles and practices
K0690	Knowledge of requirements analysis principles and practices
K0691	Knowledge of cyber defense tools and techniques
K0692	Knowledge of vulnerability assessment tools and techniques
K0694	Knowledge of computer algorithm capabilities and applications
K0698	Knowledge of cryptographic key management principles and practices
K0707	Knowledge of database systems and software
K0709	Knowledge of business continuity and disaster recovery (BCDR) policies and procedures
K0710	Knowledge of enterprise cybersecurity architecture principles and practices
K0714	Knowledge of electrical engineering principles and practices
K0718	Knowledge of network communications principles and practices
K0719	Knowledge of human-computer interaction (HCI) principles and practices
K0720	Knowledge of Security Assessment and Authorization (SA&A) processes
K0727	Knowledge of analysis standards and best practices
K0728	Knowledge of Confidentiality, Integrity and Availability (CIA) principles and practices
K0729	Knowledge of non-repudiation principles and practices
K0730	Knowledge of cyber safety principles and practices
K0731	Knowledge of systems security engineering (SSE) principles and practices
K0736	Knowledge of information technology (IT) security principles and practices
K0739	Knowledge of mathematics principles and practices
K0742	Knowledge of identity and access management (IAM) principles and practices
K0743	Knowledge of new and emerging technologies
K0744	Knowledge of operating system (OS) systems and software
K0745	Knowledge of parallel and distributed computing principles and practices
K0749	Knowledge of process engineering principles and practices
K0751	Knowledge of system threats
K0752	Knowledge of system vulnerabilities
K0753	Knowledge of remote access principles and practices
K0756	Knowledge of security management principles and practices
K0757	Knowledge of system design tools and techniques
K0758	Knowledge of server administration principles and practices
K0759	Knowledge of client and server architecture
K0765	Knowledge of software engineering principles and practices
K0768	Knowledge of automated systems analysis tools and techniques

Knowledge ID	Knowledge Statement
K0772	Knowledge of systems testing and evaluation tools and techniques
K0773	Knowledge of telecommunications principles and practices
K0778	Knowledge of enterprise information technology (IT) architecture principles and practices
K0779	Knowledge of systems engineering processes
K0791	Knowledge of defense-in-depth principles and practices
K0814	Knowledge of secure coding tools and techniques
K0835	Knowledge of risk assessment principles and practices
K0836	Knowledge of threat assessment principles and practices
K0839	Knowledge of critical infrastructure systems and software
K0840	Knowledge of hardware reverse engineering tools and techniques
K0842	Knowledge of software reverse engineering tools and techniques
K0846	Knowledge of secure software deployment principles and practices
K0847	Knowledge of secure software deployment tools and techniques
K0848	Knowledge of network systems management principles and practices
K0849	Knowledge of network systems management tools and techniques
K0851	Knowledge of reverse engineering principles and practices
K0859	Knowledge of encryption tools and techniques
K0865	Knowledge of data classification standards and best practices
K0866	Knowledge of data classification tools and techniques
K0868	Knowledge of process improvement principles and practices
K0869	Knowledge of process maturity models and frameworks
K0870	Knowledge of enterprise architecture (EA) reference models and frameworks
K0871	Knowledge of enterprise architecture (EA) principles and practices
K0872	Knowledge of service management principles and practices
K0873	Knowledge of service management standards and best practices
K0874	Knowledge of key management service (KMS) principles and practices
K0875	Knowledge of symmetric encryption principles and practices
K0876	Knowledge of key management service (KMS) key rotation policies and procedures
K0877	Knowledge of application firewall principles and practices
K0878	Knowledge of network firewall principles and practices
K0891	Knowledge of the Open Systems Interconnect (OSI) reference model
K0894	Knowledge of computer architecture principles and practices
K0906	Knowledge of multi-level security (MLS) systems and software
K0907	Knowledge of cross-domain solutions
K0915	Knowledge of network architecture principles and practices
K0917	Knowledge of Personally Identifiable Information (PII) data security standards and best practices
K0918	Knowledge of Payment Card Industry (PCI) data security standards and best practices
K0919	Knowledge of Personal Health Information (PHI) data security standards and best practices
K0921	Knowledge of program protection plan (PPP) principles and practices
K0927	Knowledge of configuration management tools and techniques
K0928	Knowledge of systems engineering principles and practices
K0933	Knowledge of N-tier architecture principles and practices
K0934	Knowledge of data classification policies and procedures
K0942	Knowledge of cryptology principles and practices
K0947	Knowledge of computer engineering principles and practices
K0948	Knowledge of embedded systems and software
K0949	Knowledge of fault tolerance tools and techniques
K0952	Knowledge of information theory principles and practices
K0983	Knowledge of computer networking principles and practices
K1014	Knowledge of network security principles and practices
K1049	Knowledge of routing protocols
K1080	Knowledge of secure software update principles and practices
K1081	Knowledge of secure firmware update principles and practices
K1086	Knowledge of design modeling
K1095	Knowledge of design methods
K1110	Knowledge of acquisition cybersecurity requirements
K1111	Knowledge of application security design principles and practices
K1120	Knowledge of Confidentiality, Integrity, Availability, Authenticity, and Non-repudiation (CIAAN) principles and practices
K1137	Knowledge of cybersecurity requirements
K1143	Knowledge of data classification levels

Knowledge ID	Knowledge Statement
K1159	Knowledge of fail-over or alternate site requirements
K1169	Knowledge of material supportability requirements
K1173	Knowledge of multilevel security requirements
K1194	Knowledge of Personally Identifiable Information (PII) attributes
K1212	Knowledge of security controls
K1216	Knowledge of service-oriented security architecture practices and principles
K1223	Knowledge of system backup requirements
K1236	Knowledge of user requirements

Skill (S) – 48

Skill ID	Skill Statement
S0141	Skill in assessing security systems designs
S0172	Skill in applying secure coding techniques
S0383	Skill in analyzing an organization's enterprise information technology architecture
S0385	Skill in communicating complex concepts
S0386	Skill in communicating verbally
S0387	Skill in communicating in writing
S0418	Skill in applying secure network architectures
S0419	Skill in designing systems
S0428	Skill in designing architectures
S0429	Skill in designing frameworks
S0430	Skill in collaborating with others
S0458	Skill in coordinating efforts between stakeholders
S0465	Skill in identifying critical infrastructure systems
S0466	Skill in identifying systems designed without security considerations
S0543	Skill in scanning for vulnerabilities
S0544	Skill in recognizing vulnerabilities
S0551	Skill in applying information technologies into proposed solutions
S0569	Skill in designing security controls
S0570	Skill in designing the integration of hardware solutions
S0571	Skill in designing the integration of software solutions
S0574	Skill in developing security system controls
S0578	Skill in evaluating security designs
S0590	Skill in building use cases
S0596	Skill in encrypting network communications
S0598	Skill in creating test plans
S0613	Skill in configuring software-based computer protection tools
S0632	Skill in designing Test and Evaluation Strategies (TES)
S0637	Skill in designing multi-level security solutions
S0638	Skill in designing cross-domain solutions
S0655	Skill in designing secure test plans
S0657	Skill in implementing Public Key Infrastructure (PKI) encryption
S0658	Skill in implementing digital signatures
S0659	Skill in applying security models
S0673	Skill in translating operational requirements into security controls
S0674	Skill in installing system and component upgrades
S0675	Skill in optimizing system performance
S0683	Skill in implementing network segregation
S0685	Skill in configuring computer protection components
S0686	Skill in performing risk assessments
S0728	Skill in preparing briefings
S0762	Skill in integrating organization objectives
S0791	Skill in presenting to an audience
S0813	Skill in identifying cybersecurity issues in external connections
S0814	Skill in identifying privacy issues in partner interconnections
S0822	Skill in collaborating with stakeholders

S0853	Skill in performing cybersecurity architecture analysis
S0880	Skill in performing security architecture analysis
S0893	Skill in performing user needs analysis

IO-WRL-002 - Database Administration

Group: Green (Internal)

Responsible for administering databases and data management systems that allow for the secure storage, query, protection, and utilization of data.

Tasks (T) - 14

Task ID	Task Statement
T0137	Maintain database management systems software
T0330	Maintain assured message delivery systems
T0422	Implement data management standards, requirements, and specifications
T0459	Implement data mining and data warehousing applications
T1020	Determine the operational and safety impacts of cybersecurity lapses
T1065	Determine data capacity requirements
T1066	Plan for anticipated changes in data capacity requirements
T1230	Maintain directory replication services
T1231	Maintain information exchanges through publish, subscribe, and alert functions
T1249	Perform backup and recovery of databases
T1297	Recommend new database technologies and architectures
T1402	Manage databases and data management systems
T1564	Install database management systems and software
T1565	Configure database management systems and software

Knowledge (K) - 57

Knowledge ID	Knowledge Statement
K0674	Knowledge of computer networking protocols
K0675	Knowledge of risk management processes
K0676	Knowledge of cybersecurity laws and regulations
K0677	Knowledge of cybersecurity policies and procedures
K0678	Knowledge of privacy laws and regulations
K0679	Knowledge of privacy policies and procedures
K0680	Knowledge of cybersecurity principles and practices
K0681	Knowledge of privacy principles and practices
K0682	Knowledge of cybersecurity threats
K0683	Knowledge of cybersecurity vulnerabilities
K0684	Knowledge of cybersecurity threat characteristics
K0685	Knowledge of access control principles and practices
K0686	Knowledge of authentication and authorization tools and techniques
K0698	Knowledge of cryptographic key management principles and practices
K0699	Knowledge of data administration policies and procedures
K0700	Knowledge of data standardization policies and procedures
K0701	Knowledge of data backup and recovery policies and procedures
K0702	Knowledge of data warehousing principles and practices
K0703	Knowledge of data mining principles and practices
K0704	Knowledge of database management system (DBMS) principles and practices
K0705	Knowledge of database query language capabilities and applications
K0706	Knowledge of database schema capabilities and applications
K0707	Knowledge of database systems and software
K0708	Knowledge of digital rights management (DRM) tools and techniques
K0716	Knowledge of host access control (HAC) systems and software
K0717	Knowledge of network access control (NAC) systems and software
K0742	Knowledge of identity and access management (IAM) principles and practices
K0744	Knowledge of operating system (OS) systems and software
K0746	Knowledge of policy-based access controls
K0747	Knowledge of Risk Adaptive (Adaptable) Access Controls (RAdAC)
K0750	Knowledge of query languages

Knowledge ID	Knowledge Statement
K0751	Knowledge of system threats
K0752	Knowledge of system vulnerabilities
K0759	Knowledge of client and server architecture
K0766	Knowledge of data asset management principles and practices
K0777	Knowledge of data storage media characteristics
K0859	Knowledge of encryption tools and techniques
K0862	Knowledge of data remediation tools and techniques
K0865	Knowledge of data classification standards and best practices
K0866	Knowledge of data classification tools and techniques
K0867	Knowledge of database application programming interfaces (APIs)
K0874	Knowledge of key management service (KMS) principles and practices
K0875	Knowledge of symmetric encryption principles and practices
K0876	Knowledge of key management service (KMS) key rotation policies and procedures
K0917	Knowledge of Personally Identifiable Information (PII) data security standards and best practices
K0918	Knowledge of Payment Card Industry (PCI) data security standards and best practices
K0919	Knowledge of Personal Health Information (PHI) data security standards and best practices
K0934	Knowledge of data classification policies and procedures
K0942	Knowledge of cryptology principles and practices
K0983	Knowledge of computer networking principles and practices
K0991	Knowledge of database administration principles and practices
K0992	Knowledge of database maintenance principles and practices
K1014	Knowledge of network security principles and practices
K1114	Knowledge of capacity management
K1121	Knowledge of configuration management
K1147	Knowledge of data integrity principles and practices
K1161	Knowledge of financial management

Skill (S) – 14

Skill ID	Skill Statement
S0045	Skill in optimizing database performance
S0545	Skill in designing data storage solutions
S0546	Skill in implementing data storage solutions
S0558	Skill in developing algorithms
S0559	Skill in performing data structure analysis
S0579	Skill in preparing reports
S0586	Skill in administering databases
S0728	Skill in preparing briefings
S0791	Skill in presenting to an audience
S0819	Skill in caching data
S0820	Skill in cataloging data
S0828	Skill in compiling data
S0835	Skill in distributing data
S0897	Skill in retrieving data

IO-WRL-005 - Systems Administration

Group: Green (Internal)

Responsible for setting up and maintaining a system or specific components of a system in adherence with organizational security policies and procedures. Includes hardware and software installation, configuration, and updates; user account management; backup and recovery management; and security control implementation.

Tasks (T) - 26

Task ID	Task Statement
T0431	Check system hardware availability, functionality, integrity, and efficiency
T0531	Troubleshoot hardware/software interface and interoperability problems
T1020	Determine the operational and safety impacts of cybersecurity lapses
T1092	Conduct functional and connectivity testing
T1130	Develop group policies and access control lists
T1140	Develop systems administration standard operating procedures
T1141	Document systems administration standard operating procedures
T1228	Maintain baseline system security
T1275	Determine the effectiveness of data redundancy and system recovery procedures
T1276	Develop data redundancy and system recovery procedures
T1277	Execute data redundancy and system recovery procedures
T1334	Produce cybersecurity instructional materials
T1500	Install systems and servers
T1501	Update systems and servers
T1502	Troubleshoot systems and servers
T1512	Perform periodic system maintenance
T1527	Define baseline system security requirements
T1530	Develop local network usage policies and procedures
T1531	Determine compliance with local network usage policies and procedures
T1569	Administer system and network user accounts
T1570	Establish system and network rights processes and procedures
T1571	Establish systems and equipment access protocols
T1578	Monitor system and server configurations
T1579	Maintain system and server configurations
T1588	Diagnose faulty system and server hardware
T1589	Repair faulty system and server hardware

Knowledge (K) - 79

Knowledge ID	Knowledge Statement
K0064	Knowledge of performance tuning tools and techniques
K0645	Knowledge of standard operating procedures (SOPs)
K0674	Knowledge of computer networking protocols
K0675	Knowledge of risk management processes
K0676	Knowledge of cybersecurity laws and regulations
K0677	Knowledge of cybersecurity policies and procedures
K0678	Knowledge of privacy laws and regulations
K0679	Knowledge of privacy policies and procedures
K0680	Knowledge of cybersecurity principles and practices
K0681	Knowledge of privacy principles and practices
K0682	Knowledge of cybersecurity threats
K0683	Knowledge of cybersecurity vulnerabilities
K0684	Knowledge of cybersecurity threat characteristics
K0685	Knowledge of access control principles and practices
K0686	Knowledge of authentication and authorization tools and techniques
K0710	Knowledge of enterprise cybersecurity architecture principles and practices
K0712	Knowledge of Local Area Networks (LAN)

Knowledge ID	Knowledge Statement
K0713	Knowledge of Wide Area Networks (WAN)
K0716	Knowledge of host access control (HAC) systems and software
K0717	Knowledge of network access control (NAC) systems and software
K0728	Knowledge of Confidentiality, Integrity and Availability (CIA) principles and practices
K0729	Knowledge of non-repudiation principles and practices
K0730	Knowledge of cyber safety principles and practices
K0731	Knowledge of systems security engineering (SSE) principles and practices
K0736	Knowledge of information technology (IT) security principles and practices
K0737	Knowledge of bandwidth management tools and techniques
K0740	Knowledge of system performance indicators
K0741	Knowledge of system availability measures
K0744	Knowledge of operating system (OS) systems and software
K0746	Knowledge of policy-based access controls
K0747	Knowledge of Risk Adaptive (Adaptable) Access Controls (RAdAC)
K0749	Knowledge of process engineering principles and practices
K0751	Knowledge of system threats
K0752	Knowledge of system vulnerabilities
K0758	Knowledge of server administration principles and practices
K0759	Knowledge of client and server architecture
K0760	Knowledge of server diagnostic tools and techniques
K0761	Knowledge of Fault Detection and Diagnostics (FDD) tools and techniques
K0765	Knowledge of software engineering principles and practices
K0770	Knowledge of system administration principles and practices
K0778	Knowledge of enterprise information technology (IT) architecture principles and practices
K0779	Knowledge of systems engineering processes
K0780	Knowledge of hardware maintenance policies and procedures
K0781	Knowledge of virtual private network (VPN) systems and software
K0791	Knowledge of defense-in-depth principles and practices
K0792	Knowledge of network configurations
K0794	Knowledge of file system implementation principles and practices
K0805	Knowledge of command-line tools and techniques
K0806	Knowledge of machine virtualization tools and techniques
K0829	Knowledge of account creation policies and procedures
K0830	Knowledge of password policies and procedures
K0837	Knowledge of hardening tools and techniques
K0840	Knowledge of hardware reverse engineering tools and techniques
K0842	Knowledge of software reverse engineering tools and techniques
K0851	Knowledge of reverse engineering principles and practices
K0858	Knowledge of virtual machine detection tools and techniques
K0859	Knowledge of encryption tools and techniques
K0870	Knowledge of enterprise architecture (EA) reference models and frameworks
K0871	Knowledge of enterprise architecture (EA) principles and practices
K0877	Knowledge of application firewall principles and practices
K0878	Knowledge of network firewall principles and practices
K0891	Knowledge of the Open Systems Interconnect (OSI) reference model
K0915	Knowledge of network architecture principles and practices
K0917	Knowledge of Personally Identifiable Information (PII) data security standards and best practices
K0918	Knowledge of Payment Card Industry (PCI) data security standards and best practices
K0919	Knowledge of Personal Health Information (PHI) data security standards and best practices
K0925	Knowledge of wireless communication tools and techniques
K0926	Knowledge of signal jamming tools and techniques
K0928	Knowledge of systems engineering principles and practices
K0935	Knowledge of incident, event, and problem management policies and procedures
K0946	Knowledge of incident reporting policies and procedures
K0947	Knowledge of computer engineering principles and practices
K0958	Knowledge of system integration principles and practices
K0983	Knowledge of computer networking principles and practices
K1014	Knowledge of network security principles and practices
K1069	Knowledge of virtual machine tools and technologies
K1072	Knowledge of automated security control testing tools and techniques

Knowledge ID	Knowledge Statement
K1188	Knowledge of organizational policies and procedures
K1226	Knowledge of systems administration standard operating policies and procedures

Skill (S) – 32

Skill ID	Skill Statement
S0045	Skill in optimizing database performance
S0111	Skill in interfacing with customers
S0407	Skill in developing standard operating procedures (SOPs)
S0408	Skill in maintaining standard operating procedures (SOPs)
S0430	Skill in collaborating with others
S0438	Skill in functioning effectively in a dynamic, fast-paced environment
S0448	Skill in creating automated security control systems
S0449	Skill in maintaining automated security control systems
S0451	Skill in deploying continuous monitoring technologies
S0472	Skill in developing virtual machines
S0473	Skill in maintaining virtual machines
S0487	Skill in operating IT systems
S0488	Skill in maintaining IT systems
S0561	Skill in configuring software
S0577	Skill in troubleshooting computer networks
S0582	Skill in troubleshooting system performance
S0587	Skill in maintaining directory services
S0593	Skill in handling incidents
S0606	Skill in manipulating operating system components
S0613	Skill in configuring software-based computer protection tools
S0662	Skill in managing servers
S0663	Skill in managing workstations
S0672	Skill in troubleshooting failed system components
S0674	Skill in installing system and component upgrades
S0675	Skill in optimizing system performance
S0677	Skill in recovering failed systems
S0678	Skill in administering operating systems
S0687	Skill in performing administrative planning activities
S0729	Skill in preparing plans
S0741	Skill in administering servers
S0762	Skill in integrating organization objectives
S0844	Skill in managing account access rights

PD-WRL-001 - Defensive Cybersecurity

Group: Green (Internal)

Responsible for analyzing data collected from various cybersecurity defense tools to mitigate risks.

Tasks (T) - 43

Task ID	Task Statement
T0020	Develop content for cyber defense tools
T0164	Perform cyber defense trend analysis and reporting
T0292	Recommend computing environment vulnerability corrections
T0299	Identify network mapping and operating system (OS) fingerprinting activities
T1020	Determine the operational and safety impacts of cybersecurity lapses
T1021	Review cyber defense service provider reporting structure
T1084	Identify anomalous network activity
T1085	Identify potential threats to network resources
T1112	Validate network alerts
T1119	Recommend vulnerability remediation strategies
T1176	Determine if cybersecurity-enabled products reduce identified risk to acceptable levels
T1177	Determine if security control technologies reduce identified risk to acceptable levels
T1241	Document cybersecurity incidents
T1242	Escalate incidents that may cause ongoing and immediate impact to the environment
T1254	Determine the effectiveness of an observed attack
T1266	Recommend risk mitigation strategies
T1278	Recommend system modifications
T1290	Communicate daily network event and activity reports
T1299	Determine causes of network alerts
T1347	Detect cybersecurity attacks and intrusions
T1348	Distinguish between benign and potentially malicious cybersecurity attacks and intrusions
T1349	Communicate cybersecurity attacks and intrusions alerts
T1350	Perform continuous monitoring of system activity
T1351	Determine impact of malicious activity on systems and information
T1384	Establish intrusion set procedures
T1386	Analyze network traffic anomalies
T1387	Validate intrusion detection system alerts
T1388	Isolate malware
T1389	Remove malware
T1390	Identify network device applications and operating systems
T1391	Reconstruct malicious attacks
T1406	Construct cyber defense network tool signatures
T1428	Notify designated managers, cyber incident responders, and cybersecurity service provider team members of suspected cybersecurity incidents
T1539	Analyze organizational cybersecurity posture trends
T1540	Develop organizational cybersecurity posture trend reports
T1541	Develop system security posture trend reports
T1548	Determine adequacy of access controls
T1582	Maintain currency of cyber defense threat conditions
T1583	Determine effectiveness of system implementation and testing processes
T1603	Recommend threat and vulnerability risk mitigation strategies
T1615	Advise stakeholders on vulnerability compliance
T1616	Resolve computer security incidents
T1618	Advise stakeholders on disaster recovery, contingency, and continuity of operations plans

Knowledge (K) - 125

Knowledge ID	Knowledge Statement
K0018	Knowledge of encryption algorithms
K0068	Knowledge of programming language structures and logic
K0674	Knowledge of computer networking protocols
K0675	Knowledge of risk management processes
K0676	Knowledge of cybersecurity laws and regulations
K0677	Knowledge of cybersecurity policies and procedures
K0678	Knowledge of privacy laws and regulations
K0679	Knowledge of privacy policies and procedures
K0680	Knowledge of cybersecurity principles and practices
K0681	Knowledge of privacy principles and practices
K0682	Knowledge of cybersecurity threats
K0683	Knowledge of cybersecurity vulnerabilities
K0684	Knowledge of cybersecurity threat characteristics
K0685	Knowledge of access control principles and practices
K0686	Knowledge of authentication and authorization tools and techniques
K0689	Knowledge of network infrastructure principles and practices
K0691	Knowledge of cyber defense tools and techniques
K0692	Knowledge of vulnerability assessment tools and techniques
K0694	Knowledge of computer algorithm capabilities and applications
K0698	Knowledge of cryptographic key management principles and practices
K0707	Knowledge of database systems and software
K0710	Knowledge of enterprise cybersecurity architecture principles and practices
K0716	Knowledge of host access control (HAC) systems and software
K0717	Knowledge of network access control (NAC) systems and software
K0718	Knowledge of network communications principles and practices
K0723	Knowledge of vulnerability data sources
K0724	Knowledge of incident response principles and practices
K0725	Knowledge of incident response tools and techniques
K0726	Knowledge of incident handling tools and techniques
K0728	Knowledge of Confidentiality, Integrity and Availability (CIA) principles and practices
K0729	Knowledge of non-repudiation principles and practices
K0730	Knowledge of cyber safety principles and practices
K0731	Knowledge of systems security engineering (SSE) principles and practices
K0732	Knowledge of intrusion detection tools and techniques
K0736	Knowledge of information technology (IT) security principles and practices
K0742	Knowledge of identity and access management (IAM) principles and practices
K0743	Knowledge of new and emerging technologies
K0744	Knowledge of operating system (OS) systems and software
K0746	Knowledge of policy-based access controls
K0747	Knowledge of Risk Adaptive (Adaptable) Access Controls (RAdAC)
K0749	Knowledge of process engineering principles and practices
K0751	Knowledge of system threats
K0752	Knowledge of system vulnerabilities
K0756	Knowledge of security management principles and practices
K0757	Knowledge of system design tools and techniques
K0758	Knowledge of server administration principles and practices
K0759	Knowledge of client and server architecture
K0765	Knowledge of software engineering principles and practices
K0766	Knowledge of data asset management principles and practices
K0770	Knowledge of system administration principles and practices
K0772	Knowledge of systems testing and evaluation tools and techniques
K0773	Knowledge of telecommunications principles and practices
K0778	Knowledge of enterprise information technology (IT) architecture principles and practices
K0779	Knowledge of systems engineering processes
K0781	Knowledge of virtual private network (VPN) systems and software
K0783	Knowledge of network attack characteristics
K0784	Knowledge of insider threat laws and regulations
K0785	Knowledge of insider threat tools and techniques

Knowledge ID	Knowledge Statement
K0788	Knowledge of adversarial tactics principles and practices
K0789	Knowledge of adversarial tactics tools and techniques
K0790	Knowledge of adversarial tactics policies and procedures
K0791	Knowledge of defense-in-depth principles and practices
K0792	Knowledge of network configurations
K0793	Knowledge of file extensions
K0805	Knowledge of command-line tools and techniques
K0812	Knowledge of digital communication systems and software
K0813	Knowledge of interpreted and compiled programming language characteristics
K0815	Knowledge of intelligence collection management processes
K0816	Knowledge of front-end intelligence collection systems and software
K0829	Knowledge of account creation policies and procedures
K0830	Knowledge of password policies and procedures
K0831	Knowledge of network attack vectors
K0832	Knowledge of cyberattack characteristics
K0833	Knowledge of cyberattack actor characteristics
K0837	Knowledge of hardening tools and techniques
K0840	Knowledge of hardware reverse engineering tools and techniques
K0842	Knowledge of software reverse engineering tools and techniques
K0844	Knowledge of cyberattack stages
K0845	Knowledge of cyber intrusion activity phases
K0848	Knowledge of network systems management principles and practices
K0849	Knowledge of network systems management tools and techniques
K0851	Knowledge of reverse engineering principles and practices
K0859	Knowledge of encryption tools and techniques
K0860	Knowledge of malware signature principles and practices
K0861	Knowledge of network port capabilities and applications
K0870	Knowledge of enterprise architecture (EA) reference models and frameworks
K0871	Knowledge of enterprise architecture (EA) principles and practices
K0877	Knowledge of application firewall principles and practices
K0878	Knowledge of network firewall principles and practices
K0879	Knowledge of industry cybersecurity models and frameworks
K0880	Knowledge of access control models and frameworks
K0891	Knowledge of the Open Systems Interconnect (OSI) reference model
K0892	Knowledge of cyber defense laws and regulations
K0915	Knowledge of network architecture principles and practices
K0917	Knowledge of Personally Identifiable Information (PII) data security standards and best practices
K0918	Knowledge of Payment Card Industry (PCI) data security standards and best practices
K0919	Knowledge of Personal Health Information (PHI) data security standards and best practices
K0924	Knowledge of network analysis tools and techniques
K0928	Knowledge of systems engineering principles and practices
K0937	Knowledge of countermeasure design principles and practices
K0938	Knowledge of network mapping principles and practices
K0939	Knowledge of packet-level analysis tools and techniques
K0940	Knowledge of subnet tools and techniques
K0942	Knowledge of cryptology principles and practices
K0947	Knowledge of computer engineering principles and practices
K0948	Knowledge of embedded systems and software
K0950	Knowledge of Intrusion Detection System (IDS) tools and techniques
K0951	Knowledge of Intrusion Prevention System (IPS) tools and techniques
K0955	Knowledge of penetration testing principles and practices
K0956	Knowledge of penetration testing tools and techniques
K0962	Knowledge of targeting laws and regulations
K0963	Knowledge of exploitation laws and regulations
K0969	Knowledge of cyberattack tools and techniques
K0983	Knowledge of computer networking principles and practices
K1014	Knowledge of network security principles and practices
K1079	Knowledge of web application security risks
K1089	Knowledge of protocol analyzer tools and techniques
K1108	Knowledge of traceroute tools and techniques

Knowledge ID	Knowledge Statement
K1131	Knowledge of cyber defense monitoring tools
K1132	Knowledge of cyber defense system analysis tools
K1144	Knowledge of data correlation tools and techniques
K1168	Knowledge of intrusion set tools and techniques
K1176	Knowledge of network topologies
K1181	Knowledge of organizational cybersecurity incident response plans
K1193	Knowledge of packet analysis tools and techniques

Skill (S) – 38

Skill ID	Skill Statement
S0156	Skill in performing packet-level analysis
S0483	Skill in identifying software communications vulnerabilities
S0490	Skill in recreating network topologies
S0509	Skill in evaluating security products
S0543	Skill in scanning for vulnerabilities
S0544	Skill in recognizing vulnerabilities
S0566	Skill in developing signatures
S0567	Skill in deploying signatures
S0572	Skill in detecting host- and network-based intrusions
S0574	Skill in developing security system controls
S0578	Skill in evaluating security designs
S0593	Skill in handling incidents
S0600	Skill in collecting relevant data from a variety of sources
S0614	Skill in categorizing types of vulnerabilities
S0627	Skill in reading signatures
S0651	Skill in performing malware analysis
S0667	Skill in assessing security controls
S0688	Skill in performing network data analysis
S0712	Skill in evaluating data source quality
S0722	Skill in interpreting traceroute results
S0755	Skill in reconstructing a network
S0809	Skill in utilizing cyber defense service provider information
S0838	Skill in identifying anomalous activities
S0839	Skill in identifying exploited system weaknesses
S0840	Skill in identifying misuse activities
S0846	Skill in monitoring system activity
S0854	Skill in performing data analysis
S0857	Skill in performing dynamic analysis
S0859	Skill in performing event correlation
S0863	Skill in performing incident analysis
S0866	Skill in performing log file analysis
S0867	Skill in performing malicious activity analysis
S0869	Skill in performing metadata analysis
S0872	Skill in performing network data flow analysis
S0874	Skill in performing network traffic analysis
S0875	Skill in performing network traffic packet analysis
S0885	Skill in performing system activity analysis
S0892	Skill in performing trend analysis